Spec Coverage — `BizzEngager - Final Draft.pdf` vs. this codebase

Every line-item in the 15-page spec, per module, checked against the current code. Audited 2026-08-05 against `master`.

Legend — ✅ implemented · ⚠️ partial (detail differs or half-built) · ❌ missing · ➕ built but not in the spec

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Part A — Superadmin Panel

A.1 Dashboard

Code: `frontend/src/pages/superadmin/Dashboard.jsx`, `backend/src/modules/superadmin/dashboard/dashboard.routes.js`

#	Spec detail	Status	Notes
1	Dynamic welcome message with the Superadmin's name	✅	Time-aware greeting from `profile.name` (`lib/greeting.js`); falls back to "Dashboard" while the profile loads.
2	Global date range filter — Last Week / Last Month / Last Year / Custom	✅	All four. Custom reveals a start/end pair seeded with the current window; fixed an inclusive-end-date off-by-one in `resolveDateWindow` while wiring it up.
3	Total Companies + growth % vs equivalent prior period	✅	`growthPct(newCompanies, prevCompanies)`.
4	Total Revenue + same period-over-period growth	✅	
5	Error logs panel	❌	**Removed on request** this session and replaced with the Revenue chart. Deliberate deviation. Data still exists (`ErrorLog`) and is visible under System Logs.
6	Revenue chart	✅	Area chart, now in the 2-up row.
7	Users/Companies activity chart	✅	"Companies Added" bar chart.
8	Total presets added by the Superadmin	✅	

A.2 Analytics & Reporting

Code: `frontend/src/pages/superadmin/Analytics.jsx`

#	Spec detail	Status	Notes
1	Revenue — ARPU	✅	revenue ÷ active companies
2	Companies — active, most active, running low on credits	✅	All three; Low on Credits is now paginated (8/page, cap 100).
3	Total Frames, broken down by user	✅	Tile + per-company table column
4	Total Broadcasts, broken down by user	✅	
5	Total Messages Sent — per user	✅	
6	Average Credit Usage — per user	✅	
7	Detailed per-user panel, every statistic for one user	✅	`PerCompanyBreakdown`, paginated + CSV export
8	Credit usage breakdown per user/company	✅	`creditsUsed` / `creditsLeft` columns

A.3 Companies

Code: `frontend/src/pages/superadmin/Companies.jsx`, `backend/src/modules/superadmin/companies/`

#	Spec detail	Status	Notes
1	Search bar	✅	

#	Spec detail	Status	Notes
2	Filters — tag, start date, end date	✅	Plus a status filter (➕)
3	"Add Company" action	✅	
4	Manage Tags popup — Tag Name, Total Companies, Date & Time; Edit/Delete	✅	`companyCount` subquery
5	Add Company fields — Tag(s) **required**, Owner Name, Company Name req, Mobile req, Email req, Address, Credits opt, GST, Password	✅	Tags enforced client-side. Owner Name is optional (matches spec — it is *not* marked required there).
6	Listing columns — Company Name, Tags, Credits (inline "+ Add"), Used Credits, Status, Email, Owner Name, Created	✅	Owner is its own sortable column; the Credits cell carries an inline "+ Add" chip. Email stays a sub-line under Company Name.
7	Row actions — Add Credit, Edit, Activate/Deactivate, Archive/Delete (restorable), View, Login-as	✅	All present. Archive is the spec's "Delete" — a confirmed soft-delete that locks the company's users out and stays restorable; archived rows show the archive date and swap their actions for View + Restore.

A.4 Channel Configuration

Code: `frontend/src/pages/superadmin/Manage.jsx` → `ChannelConfig`, `backend/.../channels/provider.routes.js`

#	Spec detail	Status	Notes
1	Three tabs — WhatsApp, SMS, Email	✅	One tab per channel; each remounts its own list.
2	Listing — Provider/Channel Name, Base URL / API Endpoint, Added/Updated Date & Time	✅	Provider (+ key), Base URL / endpoint, Rate/s, Active, and an Added / Updated column.
3	Row actions — View, Edit, Delete	✅	View opens a read-only detail modal; Edit + Delete via CrudPage.

A.5 System Logs

Code: `frontend/src/pages/superadmin/Logs.jsx`, `backend/.../superadmin/logs/logs.routes.js`

#	Spec detail	Status	Notes
1	Filters — Company Name, Start Date, End Date	✅	
2	Download Logs (Excel export)	✅	Per-tab `.xlsx`, respects the date range
3	Login Logs tab — Company, Status, Email, Date & Time	✅	"Event" badge + IP (➕)
4	Campaign Logs tab — Company, Email, Channel, Contacts, Date & Time	✅	Email column added to the table and the Excel export. The column existed on the model but was never written — the campaign runtime now fills it, and historical rows were backfilled.
5	Error Logs tab — Company, Message, Date & Time	✅	Plus Source (➕)

A.6 Transaction Logs

Code: `frontend/src/pages/superadmin/Transactions.jsx`

#	Spec detail	Status	Notes
1	Search bar	✅	
2	Filters — Company, Status, Start/End date	✅	
3	Download (Excel export)	✅	
4	Columns — Invoice ID, Company, Status, Amount, Credit, Type, Date & Time	✅	All seven

#	Spec detail	Status	Notes
5	Row actions — View (full detail), Download (PDF)	✅	Detail modal + PDF invoice

A.7 Preset & Shape Management

Flagged in the spec as the single most important module.

Code: `backend/src/modules/superadmin/library/library.routes.js` , `frontend/src/pages/superadmin/Manage.jsx` → `Library`

7.1 Presets

#	Spec detail	Status	Notes
1	Superadmin can create, edit, publish presets	✅	Built in the shared editor at `/superadmin/preset-editor`
2	A preset is a connected group of shapes forming one cohesive graphic	✅	Stored as grouped canvas JSON
3	Frames are never presets; a preset sits on top of a frame	✅	Separate `Frame` and `Preset` models
4	Text must be insertable inside every shape in a preset	✅	`shapeText` prop on shapes
5	Draft → Published → Archived workflow before visibility in company editors	✅	`/presets/:id/publish
6	Preset categories manageable without a code release	✅	`PresetCategory` CRUD, creatable inline from the editor
7	Same editor as the Admin panel	✅	`FrameEditorPage mode="preset"`
—	**Audience scoping — all companies or selected tags**	➕	Added this session; not in the spec. `audience` + `audienceTagIds` on `Preset` , filtered in `/admin/design/brand-presets` .

7.2 Shapes

#	Spec detail	Status	Notes
1	Superadmin can add shapes to the shared library	✅	
2	Direct SVG upload with the same per-region recolor pipeline as the editor	✅	`UploadSvgButton` + `recolorRegions`
3	One ingestion path for library and admin-uploaded custom shapes	✅	Shared `Shape` model, scoped brand/company
4	Same editor for Superadmin	✅	

A.8 Settings

Code: `frontend/src/pages/superadmin/SettingsTabs.jsx` , `Manage.jsx`

8.1 Profile & Security

Spec detail	Status	Notes
Editable Superadmin account details	✅	
Sign-out	✅	
2FA — email verification **and** authenticator app (TOTP)	✅	`/auth/2fa/totp/` + `/auth/2fa/email/`

8.2 Superadmin Management & Permissions

Spec detail	Status	Notes
Listing — Name, Email, Created, Last Login	✅	
Row actions — Edit, View, Delete	✅	View opens a read-only account + permission detail; Edit + Delete alongside.
Search bar + "Add New"	✅	
Add popup — Name, Email, Password, Confirm Password	✅	
Granular permissions per module	✅	All 13 spec modules present with their action sets (Companies, Billing, System Logs, Presets, Shapes, Analytics, Version, Notifications, Credit Config, Custom Fields, Channel Config…)
"Special" action beyond Add/Update/Delete	✅	Declared for all seven modules the spec lists. Enforced where a beyond-CRUD action exists: impersonation, log export, preset publish/archive/draft, and the INR→credit rate.

8.3 Version Update & Changelog

Spec detail	Status
Listing — Version ID, Name, Description	✅
View action with full changelog	✅
"Launch New Version" popup — Name, Description, Changelog	✅
Rollout target — All Companies or selected companies/tags	✅ (`rolloutTarget` JSON, resolved server-side)

8.4 Notifications

Spec detail	Status
Listing — Notification ID, Title, Description	✅
View action	✅
"Send New Notification" — Title, Description	✅
Recipient target — All Companies or selected companies/tags	✅ (+ realtime socket push, ➕)

8.5 Credit Configuration

Spec detail	Status	Notes
Element-Based Rules — cost by which elements exist in a frame	✅	`ElementRule` CRUD + live estimate in the editor
(Open question in the spec: flat / by category / scale with count)	✅	Resolved as per-element-type rules priced by count
Credit Cost Configuration — INR-to-credit rate	✅	`CreditRateTab` , `GET/PUT /superadmin/credits/config`

8.6 Custom Field Management

Spec detail	Status	Notes
Listing — Field Name, Field Type, Scope	✅	Scope column + picker: "All companies" or selected company tags (`appliesTo` / `appliesToTagIds`), enforced at render-time field resolution. "Category-wise" is implemented as company tags — the platform's segmentation primitive everywhere else. "Admin" scope = company-defined fields, managed in the company panel.
Row actions — Edit, Delete	✅	
"Add New Custom Field" popup	✅	

Part B — Admin (Company) Panel

B.1 Dashboard

Code: `frontend/src/pages/admin/Dashboard.jsx`

#	Spec detail	Status	Notes
1	Dynamic welcome message with the company name	✅	Same greeting helper, from `profile.companyName`
2	Date range filter (Week/Month/Year/Custom)	✅	Same component as A.1, Custom included
3	Total Contacts, Campaigns, Frames, Scheduled Campaigns	✅	
4	Delivered Ratio and Read Ratio	✅	
5	Channel integration status — WhatsApp / Email / SMS	✅	"Channel Integrations" card
6	Total campaign graph	✅	
7	Credit usage graph	✅	

B.2 Contacts

Code: `frontend/src/pages/admin/Contacts.jsx`

#	Spec detail	Status
1	Search bar	✅
2	Add Contact — Tags, Business Name, Owner Name, Contact Number, Email, Business Logo, per-channel consent	✅
3	Bulk Contact Upload — tag selection + CSV	✅ (with sample CSV + header alias auto-mapping)
4	Bulk archive/delete	✅
5	Manage Tags — add, update, delete	✅
6	Manage Custom Fields — Name, auto Alias, Type, Required toggle	✅
7	Filters — Tags, Date range	✅
8	Listing — Tag, Business Name, Owner Name, Mobile, Consent, Date	✅
9	Row actions — View, Edit, Archive/Delete with recovery window	✅ (`/archive` + `/restore`)
10	View Contact showing every configured field	✅
11	Opt-out link marks the contact opted-out per channel and excludes it from future campaigns	✅ (`modules/public/optout.routes.js` , honoured in audience counts)

B.3 Frames

Code: `frontend/src/pages/admin/Misc.jsx` → `Frames`

#	Spec detail	Status
1	"Add Frame" button	✅
2	Manage Categories (add/update/delete + Date & Time)	✅
3	Search bar	✅
4	Filters — Category, Date range, Aspect ratio/size	✅
5	Listing — Image, Frame Name, Category, Size, Date & Time	✅
6	Row actions — Edit, Duplicate, View, Archive/Delete with recovery	✅
7	Add Frame popup — Name, Category, Size	✅

B.3a Frame Editor

Code: `frontend/src/pages/admin/frame-editor/`

Tab order matches the spec exactly: **Arrange, Elements, Photos, Presets, Text**.

Live credit estimate is shown persistently in the toolbar ✅.

1. Arrange

Spec detail	Status	Notes
1.1 Align to Frame — L/C/R/T/M/B	✅	
1.1 Align to Selection (2+ objects)	✅	
1.1 Align to Key Object	✅	
1.1 Align to a Specific Point (X/Y)	✅	

Spec detail	Status	Notes
1.1 Switch reference point at any time	✅	Selection / Frame / Key Object / Custom
1.2 Distribute H/V, Distribute Centers H/V	✅	
1.2 Toggle within selection vs across frame	✅	
1.3 Smart spacing guides with live gap distance	✅	
1.3 Snap to grid (toggle + size)	✅	
1.3 Snap to margins / safe zones	✅	
1.3 Custom guide lines dragged off the ruler, lockable	✅	`Rulers.jsx`
1.4 Bring to Front / Forward / Backward / Send to Back	✅	
1.5 Numeric X, Y, W, H, Rotation	✅	
1.5 Lock aspect ratio toggle	✅	
1.5 Flip Horizontal / Vertical	✅	
1.6 Layer list, front-to-back, drag to reorder	✅	
1.6 Rename / Show-Hide / Lock-Unlock per layer	✅	
1.6 Group / ungroup with expand-collapse	✅	
1.6 Multi-select for batch lock/hide/delete/move	✅	
1.6 Thumbnail preview per layer	✅	
1.6 Two-way selection sync	✅	
1.6 Search/filter layers by name	✅	

2. Elements

Spec detail	Status	Notes
2.1 Shapes — rect, circle, line, arrow, polygon, custom path…	✅	10 primitives incl. star, heart, triangle, ellipse

Spec detail	Status	Notes
2.1 Icons — line / filled / duotone style	⚠️	Icon library + `style` field exist; the three-style split is thin in the seeded set.
2.1 Lines & Connectors	✅	Connectors anchor to two objects by id and re-route on move/scale/rotate; they self-remove when an anchor is deleted.
2.2 Keyword search	✅	
2.2 "Recently Used" row	✅	
2.3 Click to insert, drag-and-drop, shortcuts	✅	
2.4 Recolor vector — one swatch per region	✅	
2.4 Recolor raster — duotone, shadow + highlight	✅	Luminance mapped across a shadow→highlight ramp via stock fabric filters, so the server renderer re-applies it from the serialized JSON. Five presets plus both swatches editable.
2.4 Outline/stroke — on/off, width, colour, dash	✅	
2.4 Corner rounding — handle or exact value	✅	
2.4 Shadow/glow toggle	✅	
2.4 Flip/rotate — drag or exact degrees	✅	
2.4 Crop for frame-type/photo elements	✅	
2.4 Opacity slider	✅	
2.5 Shape combining — Combine, Weld, Trim, Intersect, Simplify	⚠️	Weld / Trim / Intersect / Simplify ✅ via paper.js. "Combine" is only distinct from Weld if it means non-destructive compound paths — currently Weld covers it.
2.6 Upload your own SVG, recolorable by region	✅	

Spec detail	Status	Notes
2.6 Draw a custom shape with a pen/path tool	✅	

3. Photos

Spec detail	Status	Notes
3.1 Upload via drag-and-drop or file browser	✅	
3.1 JPG, PNG, WEBP, HEIC (auto-converted)	✅	`normalizeUpload` does HEIC → JPEG
3.1 Search uploads by filename, delete	✅	
3.2 Unsplash — full search, featured/trending default, orientation filter, insert to canvas	✅	`unsplash.routes.js`
3.2 Unsplash is the only external source	✅	
3.3 Crop — free-form or locked to a shape	✅	
3.3 Replace image keeping position/size/crop/effects	✅	
3.3 Adjustments — brightness, contrast, saturation, temperature, sharpness	✅	All five
3.3 Preset filters with manual sliders underneath	✅	7 presets
3.3 Flip / rotate	✅	
3.3 Border/outline around the photo	✅	

4. Presets (in-editor)

Spec detail	Status
Two libraries — Brand Presets and My Presets	✅
4.1 Categories, manageable on the back end	✅
4.2 Search by name/tag, thumbnail, filter by category	✅
4.3 Drag-and-drop or click to insert	✅
4.3 Always inserted as one grouped object	✅

Spec detail	Status
4.4 Ungroup to edit, piece-by-piece	✅
4.4 Quick-edit — double-click, smart jumps ("Edit Title/Logo/Name")	✅
4.5 Save any grouping as your own preset, stored in My Presets	✅

5. Text

Spec detail	Status	Notes
5.1 Heading / Subheading / Body / blank Text Box	✅	
5.1 Font-pairing presets	✅	
5.2 Categories — Sans Serif, Serif, Display, Handwriting, Monospace	✅	
5.2 Brand Fonts pinned at the top	✅	Custom font upload
5.2 Weight selector showing only real weights	✅	`FONT_WEIGHTS` map
5.2 Font size — input, stepper, quick-pick	✅	
5.3 Bold / Italic / Underline / Strikethrough	✅	
5.3 Alignment incl. Justify	✅	
5.3 Bulleted / numbered lists	✅	
5.3 Letter spacing, line height, indent/tab stops	✅	Indent/outdent over the selected lines (or the whole box), written as leading spaces so export and per-contact rendering match the canvas exactly.
5.3 Text colour — solid or gradient	✅	
5.3 Superscript / subscript	✅	
5.3 Format Painter	✅	
5.3 Find & Replace across the document	✅	

Spec detail	Status	Notes
5.3 Vertical text orientation	✅	
5.4 Header/Footer layout	✅	
5.4 "Apply to all pages" — linked vs independent	✅	`chromeMode`
5.4 Dynamic tokens — page number, total pages, date, title	✅	`{{page}} {{pages}} {{title}}`
Shared: universal colour & fill — solid, brand swatches, wheel, hex, eyedropper	✅	
Shared: gradients — linear/radial/conic, multi-stop, per-stop opacity	✅	

B.4 Campaigns

Code: `frontend/src/pages/admin/CampaignWizard.jsx`, `Campaigns.jsx`, `backend/src/modules/admin/campaigns/`

#	Spec detail	Status	Notes
1	Listing — Frame, Campaign Name, Tag, Type, Date & Time, Status	✅	
2	Row actions — Delete, View Campaign Logs	✅	Plus resend-failed / cancel (➕)
3	Wizard: Title	✅	Step "Details"
4	Select Frame with live preview	✅	
5	Select Audience — tags + live contact count excluding opt-outs	✅	`/campaigns/audience-count`
6	Channel selection	✅	
7	Provider selection	✅	
8	Template selection — fetch & add templates inside the modal	✅	
9	Only Meta-approved WhatsApp templates selectable	✅	
10	Template variable mapping — custom value or custom field	✅	
11	Test Send to one contact / test number	✅	`POST /:id/test-send`
12	Review — full summary + credit deduction matching the editor estimate	✅	
13	Scheduling — date, time, timezone	✅	
14	Automatic rate throttling within provider limits	✅	`ratePerSecond` honoured by the BullMQ worker
15	Estimated completion time shown	✅	
16	Cannot schedule before the current time	✅	Validated

Wizard is 5 steps (Details, Frame, Audience, Provider, Review) — the spec's items are all present, folded into those steps.

B.5 Wallet

Code: `frontend/src/pages/admin/Wallet.jsx`, `backend/src/modules/admin/wallet/`

5.1 Billing

Spec detail	Status
"Add Credits" — preset packages or manual entry	✅
Search bar	✅
Download All Invoices as a ZIP of PDFs, respecting the date range	✅ (`/billing/invoices.zip`)
Columns — Invoice ID, Particulars, Type, Status, Credits, Date & Time	✅
Row action — Download Invoice (formatted PDF)	✅
Date range filter	✅
Low-balance alert threshold + auto-recharge (saved method, trigger, top-up amount)	✅ (`PUT /wallet/settings`)

5.2 Transactions

Spec detail	Status
Only deducted-credit and refunded-credit entries	✅
Search bar, date filter	✅

Spec detail	Status
Download History as Excel/CSV	✅
Columns — Transaction ID, Campaign, Type, Status, Amount (green credit / red debit), Date	✅

5.3 Logs (Activity & Logs)

Spec detail	Status
System Activity — login/logout, IP, date & time	✅
Campaign Logs — name, total contacts, Sent/Delivered/Read, status, date	✅
Error Logs — detailed message, date & time	✅

B.6 Channel Integration

Code: `frontend/src/pages/admin/Channels.jsx`, `backend/src/modules/admin/channels/`

#	Spec detail	Status	Notes
1	Three tabs — WhatsApp, SMS, Email	✅	
2	"Add Provider" button	✅	
3	WhatsApp columns — Service Provider, Account Name, Phone Number ID, Date & Time	✅	
4	Provider choice: BizzRiser, Wati, Interakt, AiSensy, DoubleTick, Zapup	✅	All six have adapters and catalogue rows. BizzRiser and Meta are bespoke; Wati, Interakt, AiSensy, DoubleTick and Zapup share `GenericBspAdapter` driven by per-provider profiles. Endpoint paths and base URLs are superadmin-overridable — see the note under Gap summary.
5	Base URL & endpoints not editable by the company	✅	Configured superadmin-side only
6	"Add New Template" with provider selection	✅	
7	Template columns — Name, Image Type, Category, Provider, Date & Time	✅	
8	Row action — Delete	✅	

B.7 Settings

Code: `frontend/src/pages/admin/Settings.jsx` , `settings/IntegrationsTabs.jsx`

7.1 Account

Spec detail	Status
Fields captured at company creation, Company ID locked	✅
Timezone	✅
Sign-out	✅
Notification history	✅

7.2 Team & Access Management

Spec detail	Status	Notes
Listing — Name, Email, Role, Date Added, Last Login	✅	
Row actions — Edit, Deactivate, Delete	✅	Deactivate/Reactivate toggle; the API already refused to deactivate the owner.
Search bar + "Add Team Member"	✅	
Add popup — Name, Email, Role with permission scope over Contacts/Frames/Campaigns/Wallet/Channel Integration	✅	8 modules covered
Role management button	✅	Roles tab
Roles listing — name, user count, added date, Edit/Delete	✅	

7.3 Security

Spec detail	Status
Change password requiring current password + email OTP	✅
2FA — email verification and TOTP	✅

7.4 Integrations

Spec detail	Status	Notes
API key	✅	Create/list/revoke

Spec detail	Status	Notes
Full documentation for every endpoint with usage examples	✅	Settings → **API Docs**. All 107 admin endpoints, grouped and searchable, each with a copy-ready curl example.
Documentation listed on the platform's Integrations page	✅	Same Settings area as API Keys and Webhooks.
—	➕	Webhooks (endpoints, event subscriptions) are built but not in the spec

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Navigation comparison

The spec calls for **eight items per panel, mirrored**. Current build:

Spec — Superadmin	Built	Spec — Admin	Built
Dashboard	✅	Dashboard	✅
Analytics & Reporting	✅ (labelled "Analytics")	Contacts	✅
Companies	✅	Frame	✅ ("Frames")
Channel Configuration	✅ ("Channel Config")	Campaign	✅ ("Campaigns")
System Logs	✅	Wallet	✅
Transaction Logs	✅ ("Transactions")	Channel Integration	✅
Presets & Shapes	✅	Activity & Logs	✅
Settings	✅	Settings	✅
—	➕ **Credits** (9th item)	—	➕ **Media**, ➕ **Templates** (10 items)

Both panels have grown past the spec's eight-item structure. Credits, Media and Templates are all reachable from within their parent modules conceptually (Settings → Credit Config, Frames → media, Channel Integration → templates), so they could be folded back if the eight-item symmetry matters.

Beyond the spec (➕)

Built and working, with no spec line-item behind it:

- **Preset audience scoping** — all companies vs. selected company tags (added this session)
- **Company invitations** — email invite flow with a public accept page (`/invite/:token`)
- **Webhooks** — endpoint registration + event subscriptions
- **Realtime notifications** — socket push to targeted companies
- **Impersonation** — superadmin "Login as company"
- **Campaign resend-failed / cancel / provider status sync**
- **Multi-page documents** in the editor with a page strip and per-page export
- **Bulk upload logs** for CSV contact imports
- **Credits nav module** — packages, element rules and conversion rate as a first-class page

Gap summary — what is actually missing

Not built at all (❌)

1. Superadmin dashboard error-logs panel (A.1) — removed on request and replaced by the revenue chart. The only remaining ❌, and a deliberate product decision rather than an omission. Re-adding it is a one-line revert whenever you want it back.

Everything else in the spec is ✅, with two judgement calls worth recording rather than "fixing":

- *Icons — line / filled / duotone style* (2.1): the library and its `style` field support all three, but the seeded icon set is thin. That is a content gap, not a code gap.
- *Shape combining — "Combine"* (2.5): Weld/Trim/Intersect/Simplify are all implemented. "Combine" is only a distinct operation if it means non-destructive compound paths; as a destructive merge it is exactly Weld, which exists.

Provider adapter caveat (B.6)

Wati, Interakt, AiSensy, DoubleTick and Zapup run through `GenericBspAdapter`, driven by a profile per provider in `backend/src/providers/whatsapp/profiles.js`. Each profile encodes that provider's published request shape — auth style, endpoint path, body layout, success detection — and all of it was exercised against a stub server (correct paths, headers, phone normalisation, variable mapping, retry classification, credential masking).

What has **not** happened is a live send against a real account for any of the five. Confirm the endpoint paths and credential names against each provider's own documentation before going live. Two mitigations are built in:

- `baseUrl` and every endpoint path are overridable per provider from **Channel Configuration**, so a correction is a settings change, not a code change.
- Wati is tenant-hosted (`live-server-<id>.wati.io`), so its base URL will almost always need overriding regardless.

Zapup is the least publicly documented of the five; its profile follows the Meta Cloud API schema that most small BSPs proxy, and should be treated as a starting point.

Verification notes

- Adapters: stub-server test covering all five profiles, plus retryable (500) vs permanent (400) classification and fail-fast on a missing credential.
- Editor: duotone matrix maths, indent round-trip and partial-selection behaviour, and connector routing/anchor-deletion checked against `fabric/node` — the same build the server-side renderer uses.
- Archive/restore, custom-field scoping and the API docs were each exercised end-to-end against the running API.